HDGMamba: High-Frequency Dynamic Guided Mamba for Robust Multispectral-Hyperspectral Image Fusion

Jiangtao Huang, Ting Luo, *Member, IEEE*, Zhouyan He, *Member, IEEE*, Haiyong Xu, *Member, IEEE*, Yang Song, and Leyuan Fang, *Senior Member, IEEE*

***Abstract*—Multispectral-hyperspectral image fusion (MHIF) generates high-resolution hyperspectral images (HRHSIs) by integrating the spectral information from low-resolution hyperspectral images (LRHSIs) with spatial details from high-resolution multispectral images (HRMSIs). Recently, researchers have focused on balancing fusion performance and computational efficiency, driving the adoption of Mamba-based architectures in MHIF. However, source images often suffer from various types of noise introduced by sensor imperfections and transmission conditions. While conventional denoising techniques can improve image quality, they often cause spatial blurring, which is an issue further amplified in Mamba due to its limited ability to model local features. To address these challenges, we propose HDGMamba, a high-frequency dynamic guided Mamba (HDGMamba) for robust MHIF tasks under noisy conditions. Its core, the high-frequency modulated state space model (HM-SSM), fuses spatial–spectral information by modeling global relationships. Crucially, HM-SSM dynamically modulates the state update process via high-frequency spatial variations to enhance fine-grained local details. Furthermore, to mitigate misleading fusion caused by noise of high-frequency features, a high-frequency gate module (HGM) is developed to extract clean high-frequency features by exploiting the distinctive gradient patterns of stochastic noise. Extensive experiments on multiple benchmark datasets demonstrate that HDGMamba significantly outperforms state-of-the-art methods in terms of both fusion accuracy and noise robustness.**

***Index Terms*—Dynamic modulation, high frequency, Mamba, multispectral-hyperspectral image fusion (MHIF).**

I. Introduction

THE hyperspectral image (HSI) accurately characterizes the spectral properties of different objects, and thus is widely applied in many fields, such as agricultural monitoring, marine surveillance, geological exploration, and so on [1], [2], [3], and [4]. However, due to the limitations of imaging devices and environmental factors, existing HSIs often suffer from low spatial resolution and significant noise [5], [6]. To enhance spatial quality, the multispectral-HSI fusion (MHIF) is used by combining the high-resolution multispectral image (HRMSI) with the low-resolution HSI (LRHSI) [7], [8]. Nevertheless, the presence of noise in both the LRHSI and the HRMSI can lead to spectral and spatial distortions during the fusion process, significantly reducing the quality of the high-resolution HSI (HRHSI). Therefore, effectively suppressing image noise while preserving spectral and spatial fidelity remains a critical challenge.

Existing MHIF methods can be broadly categorized into traditional- and deep learning-based methods [8], [9], [10]. The traditional MHIF method typically encodes prior knowledge as regularization terms by variational optimization, Bayesian inference, or nonnegative matrix factorization to solve the fusion problem [11], [12], [13], [14]. However, these methods heavily rely on manual parameter selection, thus often failing to achieve the optimal performance for all scenarios. In contrast, the deep learning-based MHIF method utilizes neural networks to learn mappings from source images (LRHSIs and HRMSIs) to target images (HRHSIs), thereby enhancing generalization capability. Sun et al. [15] utilized the wavelet transform to separate noise from LRHSIs by leveraging noise frequency characteristics, followed by convolutional neural networks (CNNs) to fuse spectral and spatial features. Dian et al. [16] adopted pretrained CNNs for denoising and iteratively optimized the reconstruction process of HRHSIs with low-rank decomposition. Although these methods improve the noise robustness of MHIF to some extent, the limited receptive field of CNNs hampers their ability to capture global contextual information in complex scenes [17]. To address this kind of limitation, Transformer architectures are employed to build long-range dependencies to enhance global modeling capabilities. However, their high computational complexity hinders practical applications [18].

In recent years, state space models (SSMs) have achieved a balance between performance and computational efficiency by capturing global information with linear-complexity [19], [20], [21]. Notably, the advanced SSM, namely Mamba, has offered new insights into MHIF research. He et al. [22] employed SSMs to construct the linear cross-attention mechanism for the HRHSI reconstruction. Peng et al. [23] leveraged the long-sequence modeling capability of SSMs to fuse spectral and

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Jiangtao Huang is with the Faculty of Information Science and Engineering, Ningbo University, Ningbo 315211, China (e-mail: 2201100032@nbu.edu.cn).

Ting Luo, Zhouyan He, and Yang Song are with the College of Science and Technology, Ningbo University, Ningbo 315212, China (e-mail: luoting@nbu.edu.cn; hezhouyan@nbu.edu.cn; songyang@nbu.edu.cn).

Haiyong Xu is with the School of Mathematics and Statistics, Ningbo University, Ningbo 315211, China (e-mail: xuhaiyong@nbu.edu.cn).

Leyuan Fang is with the School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China (e-mail: fangleyuan@gmail.com).

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spatial features from different directions. However, these methods do not adequately solve noise interference, limiting their applicability in real-world scenarios. Although the denoising techniques can be used as the preprocessing of source images, the computational cost increases [24], [25], [26]. Moreover, after denoising, the high-frequency information is lost, which causes some edge and texture regions to be overly smoothed. Critically, existing SSMs focus on exploiting global spectral–spatial correlations to benefit fusion, and their fixed state update mechanisms are difficult to adaptively capture dynamic changes in spatial details, so texture and detail recovery are affected.

To address these challenges, this article proposes a high-frequency dynamic guided Mamba (HDGMamba) for robust MHIF under complex noise conditions. Specifically, the HM-SSM is designed to model long-range dependencies for fused image reconstruction and capture high-frequency spatial variations to dynamically modulate the state update process, enhancing local details. Moreover, since spatial noise residing in high-frequency regions impacts fusion performance, an HGM is introduced to filter noise by exploiting the unstructured gradient characteristics of stochastic noise, enabling the extraction of clean high-frequency features. The experimental results demonstrate that the proposed HDGMamba outperforms existing state-of-the-art methods in terms of spectral fidelity and spatial detail recovery. The contributions of this article are summarized as follows.

1) This article proposes a novel HDGMamba for robust MHIF, namely HDGMamba. By modulating the reconstruction process of HRHSI with spatial variation cues, HDGMamba effectively fuses spectral and spatial features.
2) The HM-SSM enhances fused HRHSI quality by modeling global spectral–spatial relationships and capturing high-frequency spatial variations. While the former facilitates fused image reconstruction, the latter dynamically modulates the state update process to enhance local details.
3) The HGM is developed to separate noise from high-frequency information by leveraging structural differences, since edges and textures typically exhibit gradient continuity, while noise manifests as random pixel perturbations. This enables the extraction of clean high-frequency features to enhance the model's robustness.

The remainder of this article is organized as follows. Related work and the proposed method are introduced in Sections II and III, respectively. A detailed experimental analysis is presented in Section IV. Finally, conclusions are drawn in Section V.

II. Related Work

A. MHIF Methods

As a key technique for enhancing the spatial resolution of the HSI, MHIF has gained significant attention. MHIF primarily includes traditional- and deep learning-based methods. In particular, traditional methods are mainly divided into matrix decomposition- and tensor decomposition-based methods [12], [27]. In general, matrix decomposition-based methods decompose LRHSIs and HRMSIs into multiple matrices to extract spectral and spatial features separately, for effective fusion. Akhtar et al. [28] learned a spectral dictionary from LRHSIs and estimated the corresponding coefficient matrix from HRMSIs using Bayesian sparse coding to reconstruct HRHSIs. In contrast, tensor decomposition-based methods treat the source images as high-order tensors to preserve their spatial and spectral structural information. Li et al. [29] proposed a coupled sparse Tucker decomposition method, which simultaneously estimates the core tensor and mode dictionaries using proximal alternating optimization. However, such methods heavily rely on image priors and are sensitive to the selection of handcrafted features, thereby limiting their fusion performance.

Deep learning-based MHIF methods enhance fusion performance by learning effective representations from source images. These methods are mainly categorized into CNN- and Transformer-based approaches [30], [31], [32], [33]. Masi et al. [34] introduced a pan-sharpening neural network based on CNN to learn mappings between source and target images, thereby improving fused image quality. Yang et al. [35] proposed a dual-branch CNN to separately extract spectral and spatial features from LRHSIs and HRMSIs for more effective fusion. Since noise often exists in real-world LRHSIs, Dian et al. [16] combined a pretrained CNN denoiser into a low-rank subspace model to remove noise, and iteratively optimized the fused HRHSIs to enhance robustness. Similarly, to enhance the ability of the noise resistance, Sun et al. [15] fused spectral and spatial information in the discrete wavelet transform (DWT) domain, and employed the frequency distribution of noise to improve the quality of HRHSIs. However, due to the limited receptive field of convolutional kernels, CNN-based methods are hard to capture global information, which may result in spatial distortions. In contrast, Transformer-based methods construct the global correlations between arbitrary pixels through self-attention mechanisms, for capturing long-range dependencies [36], [37]. Deng et al. [38] proposed a pyramid shuffle-and-reshuffle Transformer (PSRT) to enhance fused image quality by capturing correlations between spectral and spatial features across different regions. Wang et al. [39] further improved spatial representation through multiscale fusion and spectral–spatial cross-attention mechanisms. In addition, Ma et al. [40] introduced a dual cross Transformer (DCTFormer) to promote interactions between spectral and spatial features for increasing the fusion quality as well. However, those methods require high computational complexity to establish long-range dependencies, thereby limiting their applicability in resource-constrained scenarios. Furthermore, several studies have integrated diffusion models into the fusion task to enhance reconstruction fidelity. For instance, Zhu et al. [41] proposed a self-supervised adaptive residual-guided subspace diffusion model (AGRS-Diff), which reconstructs HRHSI via dual-branch diffusion learning and residual-based refinement. Meanwhile, Huang et al. [42] incorporated Gaussian filters into the diffusion process to improve fusion robustness under noisy conditions. However, the inherently iterative inference process of diffusion models leads to a significant increase in computational complexity, which limits their practical efficiency in real-world scenarios. Besides combining the advantages of various network architectures, integrating models with frequency-domain features has become an important research direction for enhancing image quality. Cao et al. [43] proposed a Fourier-enhanced implicit network which improves high-frequency reconstruction by optimizing phase alignment in the frequency domain. Meanwhile, Zhou et al. [44] designed a spatial-frequency interaction network that performs multimodal fusion collaboratively in both domains.

B. State Space Model

SSMs, originally developed in control theory for modeling dynamic systems, have recently been adopted in deep learning due to their computational efficiency [45], [46], [47], [48]. Gu and Dao [49] introduced the structured SSM (S4) to capture long-range sequence dependencies with linear complexity, thus substantially decreasing computational costs in language tasks. However, the static parameters in S4 restrict its dynamic modeling capabilities. To address this, Zhu et al. [50] proposed the selective structured SSM (S6), that is Mamba, which dynamically extracts features via the selection mechanism for efficiently capturing global context.

After Mamba achieved success in natural language tasks, researchers applied it to the vision field. Liu et al. [51] introduced vision Mamba, which transforms image data into ordered visual sequences via bidirectional scanning for capturing global features. Zhu et al. [52] further enhanced spatial modeling by incorporating horizontal and vertical scanning strategies. In remote sensing image fusion, the Mamba architecture has also shown great advantages. He et al. [22] introduced PanMamba, which integrates channel-swapping mechanisms with the dynamic modeling capability of SSM to improve spectral–spatial interaction. Peng et al. [23] designed FusionMamba, which utilizes dual spectral and spatial branches to extract features from LRHSIs and HRMSIs, and employs cross Mamba to bridge the gap between them. In addition, to further improve the quality of fused images, Cao et al. [53] introduced a Mamba-based collaborative implicit neural representation fusion network (MCIFNet) to leverage Mamba for capturing global features and strengthening feature expression. Although Mamba-based methods can achieve dynamic global feature modeling with reduced computational complexity, they are hard to represent high-frequency features such as edges and textures, affecting the overall quality. Moreover, these methods inadequately solve the noise present in LRHSIs and HRMSIs, thus leading to compromised fused results in real-world scenarios.

III. Proposed Method

A. Preliminaries

SSM is a dynamic system widely used for linear time-invariant modeling. Its core mechanism involves mapping an input sequence *x(t)* to an output sequence *y(t)* via a hidden state *h(t)*. The mathematical equation of SSM is defined as

$$\begin{aligned} h'(t) &= A \cdot h(t) + B \cdot x(t) \\ y(t) &= C \cdot h(t) + D \cdot x(t) \end{aligned} \tag{1}$$

where $h'(t)$ is the first derivation of *h(t)*, representing its instantaneous rate of change. The state transition matrix $A \in \mathbb{R}^{N\times N}$ and input matrix $B \in \mathbb{R}^{1\times N}$ govern the dynamic evolution process of *h(t)*, while output matrix $C \in \mathbb{R}^{1\times N}$ and feed forward matrix $D \in \mathbb{R}^{1\times N}$ establish the mapping from input to output.

To adopt SSM for the field of computer vision, (1) is discretized using a zero-order hold. Specifically, A and B are converted to the discretized parameter matrices $\overline{A}$ and $\overline{B}$ with the timescale parameter Δ

$$\begin{aligned} \overline{A} &= \exp(\Delta \cdot A) \\ \overline{B} &= (\Delta \cdot A)^{-1} \cdot \big(\exp(\Delta \cdot A) - I\big) \cdot \Delta \cdot B \end{aligned} \tag{2}$$

where $\exp(\cdot)$ and I denote the matrix exponential operation and identity matrix, respectively. Following discretization, (1) is reformulated as:

$$\begin{aligned} h_t &= \overline{A} \cdot h_{t-1} + \overline{B} \cdot x_t \\ y_t &= C \cdot h_t + D \cdot x_t. \end{aligned} \tag{3}$$

Computational efficiency is significantly improved by transforming the iterative process in (3) into a global convolution operation. For an input sequence $x_m \in \mathbb{R}^{1\times m}$, the output sequence $y_m \in \mathbb{R}^{1\times m}$ is computed as

$$\begin{aligned} \overline{K} &= \left(D + C \cdot B \cdot C \cdot A, \ldots, C \cdot \overline{A}^{m-1} \cdot \overline{B}\right) \\ y_m &= x_m \cdot \overline{K} = x_m \cdot D + \sum_{k=0}^{m} x_{m\text{-}k} \left(C \cdot \overline{A}^{k} \cdot \overline{B}\right) \end{aligned} \tag{4}$$

where $\overline{K}$ is the global convolution kernel and $*$ denotes the convolution operation.

However, the above SSM suffers from limited representation due to the static parameter matrices. To this end, Mamba introduces a selection mechanism that enables input-dependent parameterization of B, C, and Δ

$$B, C, \Delta = \xi(x(t)) \tag{5}$$

where $\xi(\cdot)$ denotes a linear projection layer. The input-driven dynamic modeling ability of Mamba enhances cross-sequence interactions, and the proposed method achieves high-quality HRHSI reconstruction with low computational complexity by leveraging the strengths.

B. Motivation

Since Mamba models long-range dependencies efficiently with linear complexity via its continuous system and recurrent scanning mechanism, it overcomes the limited receptive fields of CNNs and the high computational complexity of Transformers. However, its underlying process of low-rank compression and temporal decay in hidden states is inherently lossy, which tends to smooth out or discard high-frequency details. In remote sensing images containing diverse land

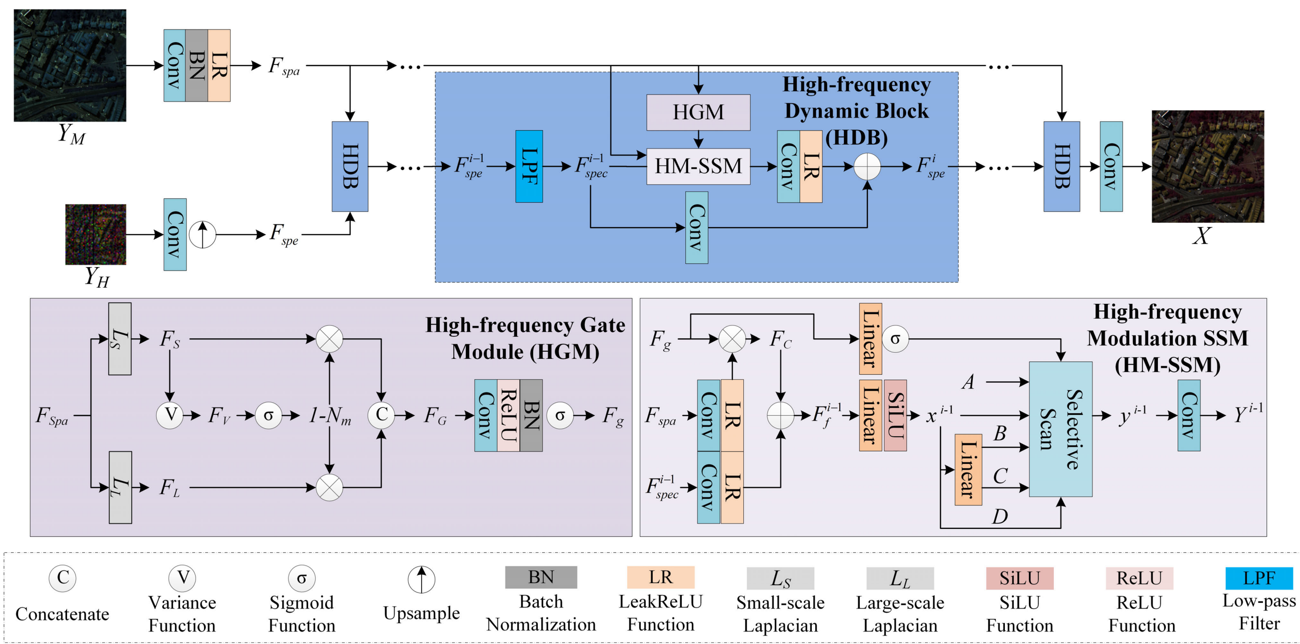

Fig. 1. Overview of HDGMamba.

cover and complex structures, fine details like object edges and textures are essential for accurate interpretation. Existing Mamba-based fusion methods typically explore spectral and spatial correlations by using multidirectional scanning strategies or optimizing input sequences. For example, PanMamba [22] reduced modal discrepancies by exchanging spectral and spatial features during fusion, while FusionMamba [23] dynamically adjusted parameter matrices by leveraging intermodal differences during the state evolution. Nevertheless, these methods generally apply a globally uniform scanning strategy across all image regions, overlooking the differing needs of detail-rich areas versus homogeneous flat regions.

To address this issue, we propose an HM-SSM, which enhances detail representation by integrating fine spatial features into the selective scanning mechanism. Specifically, since edge and texture information are concentrated primarily in the high-frequency components of HRMSI, HM-SSM adaptively adjusts its feature selection mechanism under the guidance of high-frequency cues, enabling differentiated modeling across various regions. Unlike previous local feature-enhancement methods, which typically rely on static or implicit operators like attention mechanisms or multiscale CNNs with fixed receptive fields, HM-SSM introduces a distinct strategy, which integrates high-frequency features as dynamic selection signals directly into Mamba's selective scanning process. This enables feature propagation to be adaptively modulated according to local detail richness, effectively shifting from a uniform global scan to a detail-guided and locally-aware modeling process that preserves fine spatial information.

While enhancing local detail preservation is crucial, real-world remote sensing scenarios present an additional challenge, i.e., both LRHSIs and HRMSIs are often contaminated by noise with uneven and hard-to-model distributions. Specifically, factors such as limited energy absorption in narrow spectral bands, damaged device components, or unstable transmission environments introduce various types of noise into LRHSIs. Meanwhile, HRMSIs suffer from electronic noise due to sensor constraints. Although integrating denoising techniques with fusion methods can improve the fused image quality, limitations still exist: 1) additional denoising significantly increases computational complexity and error accumulation; 2) denoising can lead to blurring spatial edges and textures, thus degrading fused image quality. To solve these drawbacks, we design the HGM to effectively isolate noise in high-frequency regions. Since spatial information exhibits regular gradient continuity while noise manifests discrete randomness, the HGM extracts clean high-frequency features via structural differences instead of smoothing, thereby preserving critical edges and textures. Through the synergistic collaboration between HM-SSM and HGM, the quality of the fused image is significantly improved.

C. High-Frequency Dynamic Guided Mamba

As illustrated in Fig. 1, we propose HDGMamba, comprising N high-frequency dynamic blocks (HDBs). Each HDB consists of an HGM, a low-pass filter (LPF), and an HM-SSM, designed to perform anti-noise fusion of LRHSI $Y_H \in \mathbb{R}^{h\times w\times C}$ and HRMSI $Y_M \in \mathbb{R}^{H\times W\times c}$ under noise scenarios while simultaneously preserving spatial details and enhancing robustness. Where, $h \times w \times C$ and $H \times W \times c$ represent the size of Y_H and Y_M, respectively.

HDGMamba includes two stages for feature extraction and feature fusion, respectively. During feature extraction, Y_H is processed via upsampling followed by convolution to obtain spectral features $F_{spe} \in \mathbb{R}^{H\times W\times C}$. Concurrently, Y_M undergoes convolutional processing to derive spatial features $F_{spa} \in \mathbb{R}^{H\times W\times c}$. The processing can be expressed as

$$\begin{aligned} F_{spe} &= \uparrow_4 (f_{conv}(Y_H)) \\ F_{spa} &= \phi(\psi(f_{conv}(Y_M))) \end{aligned} \tag{6}$$

where $\uparrow_4(\cdot)$ denotes upsampling with factor 4 while $f_{conv}(\cdot)$ is a convolution layer. Moreover, $\phi(\cdot)$ and $\psi(\cdot)$ represent LeakyReLU and batch normalization, respectively.

In the feature fusion stage, the spectral and spatial features are progressively fused by N cascaded HDBs. Each block incorporates an HGM and an HM-SMM to enhance the local detailed features, where HM-SMM fuses the spectral and spatial features via the clean high-frequency features that are extracted by HGM.

Specifically, HGM is applied to obtain the clean high-frequency features F_g from $F_{spe} \in \mathbb{R}^{H\times W\times C'}$, where C' represents the channel number of F_g. Meanwhile, the denoised spectral representation is gained as the LPF removes noise from spectral features. For instance, the output of the $(i-1)$th HDB is represented as $F_{spec}^{i-1} \in \mathbb{R}^{H\times W\times C}$ for $i \in (1, N]$. Under the guidance of F_g, HM-SMM integrates the spatial and spectral information to generate the fused feature $Y^{i-1} \in \mathbb{R}^{H\times W\times C'}$. The processing can be expressed as

$$\begin{aligned} F_g &= f_{HGM}\left(F_{spa}\right) \\ F_{spec}^{i-1} &= f_L\left(F_{spe}^{i-1}\right) \\ Y^{i-1} &= f_{HMSSM}\left(F_{spec}^{i-1}, F_g, F_{spa}\right) \end{aligned} \tag{7}$$

where $f_{HGM}(\cdot)$, $f_L(\cdot)$, and $f_{HMSSM}(\cdot)$ denote the operation of HGM, LPF, and HM-SMM, respectively.

Finally, Y^{i-1} is refined through a convolutional layer and a residual connection within the HDB, producing the output feature F_{spe}^{i}. The processing can be expressed as

$$F_{spe}^{i} = \phi\left(f_{conv}\left(Y^{i-1}\right)\right) + f_{conv}\left(F_{spec}^{i-1}\right). \tag{8}$$

During the training, L_1 loss is utilized to constrain the difference between HRHSI $X \in \mathbb{R}^{H\times W\times C}$ and ground truth $X_{gt} \in \mathbb{R}^{H\times W\times C}$. The loss function ρ can be defined as

$$\rho = \frac{1}{H\times W\times C}\sum\left|X - X_{gt}\right|. \tag{9}$$

D. High-Frequency Gate Module

Leveraging the property that edges and textures have continuous gradients while noise is randomly discrete, we propose HGM to isolate noise and extract clean high-frequency features F_g, as shown in Fig. 1.

First, HGM obtains the local detailed gradient F_S and the global structural F_L using Laplacian operators with different kernel sizes

$$\begin{aligned} F_S &= L_S\left(F_{spa}\right) \\ F_L &= L_L\left(F_{spa}\right) \end{aligned} \tag{10}$$

where $L_S(\cdot)$ and $L_L(\cdot)$ donate Laplacian operators with 3×3 and 5×5 kernel, respectively.

To identify noise locations, the noise mask N_m is computed based on the local variance of F_S

$$\begin{aligned} F_V &= V\left(F_S\right) \\ N_m &= \sigma\left(F_V\right) \end{aligned} \tag{11}$$

where $F_V(\cdot)$ represents the local variance, while $V(\cdot)$ and $\sigma(\cdot)$ are utilized to compute local variance and donate the Sigmoid function, respectively.

In high-frequency regions, noise corresponds to high gradient variance due to its irregular distribution, while edges and textures exhibit low variance owing to gradient continuity. Thus, HGM utilizes N_m to suppress noise and preserves high-frequency details, yielding denoised gradient features $F_G \in \mathbb{R}^{H\times W\times 2c}$

$$F_G = Concat\left(F_S\cdot(1-N_m), F_L\cdot(1-N_m)\right) \tag{12}$$

where $Concat(\cdot)$ denotes channelwise concatenation operation. Finally, F_G is processed through a convolutional layer, a batch normalization, and a Sigmoid function to generate F_g

$$F_g = \sigma\left(\psi\left(f_{conv}\left(F_G\right)\right)\right). \tag{13}$$

E. High-Frequency Modulation Space State Model (HM-SSM)

To enhance the ability of SSM in perceiving local information variations, we propose HM-SSM, which introduces clean high-frequency features F_g to optimize the update state process, enabling the timescale parameter to adaptively adjust parameter matrices via spatial distributions of different areas. The structure of HM-SSM is illustrated in Fig. 1.

For example, the HM-SSM in $(i-1)$th HDB process features to generate the input sequence $x^{i-1} \in \mathbb{R}^{L\times C}$, where $L = H\times W$. Specifically, both F_{spa} and F_{spec}^{i-1} through a convolution layer and a LeakRelu function to match channel dimensions. To mitigate noise in spatial features, Fspa is modulated to generate $F_C \in \mathbb{R}^{H\times W\times C'}$ via F_g, and the fused features $F_f^{i-1} \in \mathbb{R}^{H\times W\times C'}$ are generated by matrix addition between F_C and F_{spec}^{i-1}. Finally, x^{i-1} is obtained by processing F_f^{i-1} through a linear layer and SiLU function. The processing can be expressed as

$$\begin{aligned} F_C &= f_{conv}\left(\phi\left(F_{spa}\right)\right)\cdot F_g \\ F_f^{i-1} &= f_{conv}\left(\phi\left(F_{spec}^{i-1}\right)\right) + F_C \\ x^{i-1} &= \Phi\left(\xi\left(F_f^{i-1}\right)\right) \end{aligned} \tag{14}$$

where $\Phi(\cdot)$ represents SiLU function.

In HM-SSM, the parameter matrices B, C are dynamically adjusted depending on x^{i-1}, while the timescale parameter Δ is jointly modulated by x^{i-1} and F_g. The processing can be expressed as

$$\begin{aligned} B, C &= \xi\left(x^{i-1}\right) \\ \Delta &= \xi\left(x^{i-1}\right)\cdot\sigma\left(\xi\left(F_g\right)\right) = \left\{\Delta_p \mid \forall p \in [1, L]\right\}. \end{aligned} \tag{15}$$

Consequently, the discretized parameter matrices are dynamically readjusted due to the change of Δ, enabling HM-SSM to effectively capture the spatial variation. The processing can be expressed as

$$\begin{aligned} \overline{A}_p &= \exp\left(\Delta_p\cdot A\right) \\ \overline{B}_p &= \left(\Delta_p\cdot A\right)^{-1}\cdot\left(\exp\left(\Delta_p\cdot A\right) - I\right)\cdot\Delta_p\cdot B. \end{aligned} \tag{16}$$

To further enhance the modeling ability of HM-SSM, four scanning paths are adopted to obtain the multidirectional information, and the final output is generated by fusing these scanning results of different directions. The processing can be expressed as

$$y_u^{i-1} = Scan\left(x_u^{i-1}\right) u \in [1,4],$$
$$Y^{i-1} = f_{conv}\left(Concat\left(y_1^{i-1}, y_2^{i-1}, y_3^{i-1}, y_4^{i-1}\right)\right) \tag{17}$$

where $Scan(\cdot)$ represents the selective scan operation of HM-SSM, and $u \in [1,2,3,4]$ denotes the forward, backward, horizontal, and vertical paths of scanning, respectively. For example, the computation of $u = 1$ can be formulated as

$$\begin{aligned}\overline{K} &= \left(D + C \cdot \overline{B}_1, C \cdot \overline{B}_2 \cdot \overline{A}_1, \ldots, C \cdot \overline{B}_t \cdot \prod_{q=1}^{t-1} \overline{A}_q\right)\\ &\quad t \in [1, L]\\ y_1^{i-1} &= \left\{ y_{1,t}^{i-1} \mid \forall t \in [1, L]\right\}\\ y_{1,t}^{i-1} &= x_1^{i-1} \cdot D + \sum_{k=1}^{t} x_{1,t\text{-}k}^{i-1} \cdot \left(C \cdot \overline{B}_k \cdot \prod_{q=1}^{k} \overline{A}_q\right), \; t \in [1, L].\end{aligned} \tag{18}$$

Through multidirectional scanning, HM-SSM captures spatial dependencies across diverse orientations, thereby significantly enhancing its capacity for modeling fine-grained features.

IV. Experiments

A. Experimental Settings

1) Datasets: To validate the effectiveness of the proposed method, experiments are conducted on four HSI datasets, i.e., Pavia Center, Chikusei, Washington DC Mall (WDC), and Tuscany. The Pavia Center dataset was captured in 2001 by the reflective optics system imaging spectrometer (ROSIS) sensor over central Pavia, Italy. It comprises 1096×1096 pixels with 102 spectral bands, after removing water vapor absorption and noisy bands from the original 115 bands. For training, a top region of the dataset is selected, from which overlapping 64×64 pixel patches are extracted as ground truth. The corresponding HRMSI patches of size $64 \times 64 \times 3$ and LRHSI patches of size $16 \times 16 \times 102$ are generated through spatial and spectral degradation of the ground truth, respectively. The test set includes four nonoverlapping 256×256 pixel patches sampled from the remaining area of the image. The Chikusei dataset, covering agricultural and urban areas in Chikusei, Japan, contains an image of 2517×2335 pixels with 128 spectral bands spanning 363–1018 nm. The top-left regions are selected, including 1000×2200 pixels, for training, extracting overlapping 64×64 pixel patches as ground truth. HRMSI patches of size $64 \times 64 \times 3$ and LRHSI patches of size $16 \times 16 \times 128$ are derived by applying spatial and spectral degradation to the ground truth, respectively. For testing, six nonoverlapping image patches of size 680×680 pixels and 128 bands are extracted from the remaining area of the image. The WDC dataset acquires hyperspectral imagery using a hyperspectral digital imagery collection experiment (HYDICE) sensor. The full scene measures 1208×307 pixels with 210 spectral bands covering 400–2400 nm. After removing noise-affected bands, 191 spectral channels are retained. For testing, four nonoverlapping $128 \times 128 \times 191$ patches serve as ground truth. Corresponding LRHSI patches $32 \times 32 \times 191$ and HRMSI patches $128 \times 128 \times 13$ are generated through spatial and spectral degradation of the ground truth, respectively. The remaining areas form 921 overlapping $64 \times 64 \times 191$ training patches. The Tuscany dataset was acquired over Tuscany, Italy, by the PRecursore IperSpettrale della Missione Applicativa (PRISMA) hyperspectral satellite in 2024. The original image comprises 1000×1000 pixels with 234 bands. After removing damaged bands, 191 spectral bands covering the 400–2500-nm range are retained. From this dataset, two nonoverlapping $256 \times 256 \times 191$ HSI patches are selected as ground truth for testing. Corresponding LRHSI patches of size $64 \times 64 \times 191$ and HRMSI patches of size $256 \times 256 \times 10$ are generated through spatial and spectral degradation. The remaining areas are cropped into 700 overlapping $64 \times 64 \times 191$ patches for training, following the same degradation procedure as used for the test data. It is noteworthy that the Tuscany dataset uniquely provides data with real sensor noise alongside the atmospherically corrected versions of the same scenes, with the latter serving as the clean ground truth.

2) Training Settings and Evaluation Metrics: HDGMamba is implemented in PyTorch 1.1 and trained on a GeForce RTX 4090 GPU with a batch size of 16 across 400 epochs. The Adam optimizer starts with a learning rate of 5×10^{-4}, which is halved every 200 epochs. The number of feature channels is set to 64 for the Pavia Center, WDC, and Tuscany datasets, while it is set to 32 for the Chikusei dataset. Standard Gaussian noise is added to both datasets to simulate a noisy environment, with signal-to-noise ratios (SNRs) of 10 dB for LRHSIs and 35 dB for HRMSIs.

To assess the fusion performance, we employ five evaluation metrics, namely peak SNR (PSNR), structural similarity index measure (SSIM), spectral angle mapper (SAM), relative dimensionless global error in synthesis (ERGAS), and cross correlation (CC). PSNR and SSIM measure spatial similarity between the fused image and ground truth based on mean squared error (mse) and structural consistency, respectively. SAM and CC assess the spectral fidelity and spatial correlation, respectively, and ERGAS quantifies the overall error. For PSNR, SSIM, and CC, the higher values represent greater fusion performance, while the ideal values of ERGAS and SAM are zero.

3) Compared Methods: To evaluate the effectiveness of the proposed HDGMamba, eight DL-based fusion methods are compared, including one CNN-based method (CNN-FUS [16]), one diffusion-based method (RoFDiff), two transformer-based methods (PSRT [38] and DCTFormer [40]), and three mamba-based methods (PMamba [22], FMamba [23], RoFDiff [42], and AHMNet [53]).

To address the noise susceptibility of PSRT, DCT-Former, PMamba, and FMamba, three denoised methods are adopted to generate clear LRHSIs and HRMSIs, which are spectral–spatial transformer (SST [24]), spectral enhanced rectangle transformer (SERT [25]), and hyperspectral denoising transformer using low-rank prompts (Hylora [26]), respectively. Combining the four fusion methods with the

TABLE I

QUANTITATIVE ASSESSMENT OF TRAINED NOISE ON PAVIA CENTER DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE MARKED IN RED

Models	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑
CNN-FUS [16]	35.123	0.995	7.330	0.857	0.974
SST [24]+PSRT [38]	40.431	0.992	2.703	0.484	0.992
SERT [25]+PSRT	41.849	0.995	2.361	0.401	0.994
Hylora [26]+PSRT	41.437	0.994	2.553	0.419	0.994
SST+DCTFormer [40]	40.066	0.993	2.630	0.451	0.995
SERT+DCTFormer	41.774	0.995	2.505	0.372	0.995
Hylora+DCTFormer	41.482	0.995	2.574	0.384	0.995
SST+PMamba [22]	41.341	0.995	3.128	0.397	0.995
SERT+PMamba	42.193	0.996	2.745	0.358	0.996
Hylora+PMamba	41.716	0.995	3.054	0.378	0.995
SST+FMamba [23]	43.354	**0.997**	2.414	0.323	0.996
SERT+FMamba	44.174	**0.997**	2.186	0.295	**0.997**
Hylora+FMamba	43.842	**0.997**	2.328	0.306	**0.997**
RoFDiff [42]	42.860	0.996	2.049	0.319	0.996
AHMNet [53]	44.370	**0.997**	**2.057**	0.290	**0.997**
HDGMamba	**44.434**	**0.997**	2.109	**0.288**	**0.997**

three denoising methods, 12 methods are utilized to compare, which are SST + PSRT, SERT + PSRT, Hylora + PSRT, SST + DCTFormer, SERT + DCTFormer, Hylora + DCTFormer, SST + PMamba, SERT + PMamba, Hylora + PMamba, SST + FMamba, SERT + FMamba, and Hylora + FMamba, respectively. CNN-FUS, RoFDiff, and AHMNet are compared directly due to their noise robustness. All methods are trained using the configurations from their original papers.

4) Simulation Noise for Testing: LRHSIs and HRMSIs are often affected by noise due to environmental interference and sensor limitations, with LRHSIs showing higher noise intensity. Since the original datasets are noise-free, we introduce six types of complex noise into each dataset to simulate realistic conditions. For LRHSIs, the added noise includes Gaussian noise, deadline noise, impulse noise, and stripe noise, reflecting their higher vulnerability.

These simulations facilitate robustness testing of various methods under diverse noise scenarios, including: Case 1 (LRHSI [Gaussian noise (5 dB)], HRMSI [Gaussian noise (30 dB)]); Case 2 (LRHSI [Gaussian noise (5 dB) + deadline noise], HRMSI [Gaussian noise (30 dB)]); Case 3 (LRHSI [deadline noise + impulse noise], HRMSI [Gaussian noise (30 dB)]); Case 4 (LRHSI [impulse noise + stripe noise], HRMSI [Gaussian noise (30 dB)]); Case 5 (LRHSI [deadline noise + impulse noise + stripe noise], HRMSI [Gaussian noise (30 dB)]); and Case 6 (LRHSI [Gaussian noise (5 dB) + deadline noise + impulse noise + stripe noise], HRMSI [Gaussian noise (25 dB)]).

LRHSI [Gaussian noise (5 dB)] refers to zero-mean Gaussian noise added to all LRHSI bands at an SNR of 5 dB. For HRMSI [Gaussian noise (30 dB)], Gaussian noise reaches 30-dB SNR across all HRMSI bands. Deadline noise is simulated by randomly corrupting 2%–5% of column lines in each band. Impulse noise is introduced by randomly selecting 1% of pixels across all bands. Stripe noise is applied by randomly selecting 2%–5% of column lines to all bands.

B. Experimental Results

1) Comparison Results on Pavia Center Dataset: Table I presents the fusion results of the different models under trained noise, where the best results are highlighted in bold, and the second-best results are marked in red. It is clearly shown that the proposed HDGMamba achieves the best overall performance. Compared to AHMNet, the second-best method, HDGMamba achieves a higher PSNR by 0.027 dB while reducing ERGAS by 0.002. It is primarily attributed to the designed LDF and HGM, which effectively remove noise from both LRHSIs and HRMSIs. Furthermore, HDGMamba utilizes the extracted high-frequency features to guide the state evolution process in the HM-SSM, enabling more accurate reconstruction of HRHSIs and thereby significantly enhancing the final fusion quality.

As shown in Table II, presenting the fusion results under six types of untrained noise, HDGMamba consistently achieves optimal performance across most metrics in all noise conditions. Although its SAM is slightly higher than that of RoFDiff, HDGMamba maintains a clear advantage in all other quantitative indicators. This is particularly evident in Case 3, where HDGMamba surpasses RoFDiff by a substantial margin of 1.042 dB in PSNR. Overall, the comprehensive comparison confirms the superior performance of the proposed method.

Fig. 2 provides visual results and the SAM error map for Case 6, and the visual comparison further supports this conclusion. Enlarged red and yellow boxes in Fig. 2 reveal distortions across all compared methods. Specifically, Fig. 2(a)–(c), (e), (h), and (i) shows varying degrees of color distortion in the yellow box, while all compared methods suffer from detail degradation in the red box, including rough textures and visible stripe noise. In contrast, the visualization of HDGMamba displays complete detail textures, which closely match the ground truth. While the visual images of Fig. 2(n) (RoFDiff) and Fig. 2(o) (AHMNet) do not show obvious spatial distortion, their corresponding SAM error maps reveal notable spectral inaccuracies. In contrast, the SAM error map of HDGMamba is predominantly deep blue, indicating minimal spectral distortion and confirming its high spectral fidelity. This visual evidence aligns with the quantitative results, demonstrating that HDGMamba delivers superior image quality.

2) Comparison Results on Chikusei Dataset: To evaluate the generalization capability, the Chikusei dataset is also employed to assess the performance of various methods. As shown in Table III, the proposed HDGMamba achieves optimal results across all evaluation metrics under trained noise conditions. Specifically, compared to AHMNet, the second-best method, HDGMamba improves PSNR, SSIM, and CC by 0.807, 0.004, and 0.006 dB, respectively, while simultaneously reducing SAM and ERGAS by 0.064 and 0.078. The performance trends are further illustrated in Fig. 3, which visually summarizes the results across six untrained noise types. The proposed HDGMamba consistently outperforms all compared methods in this comprehensive comparison. It is worth noting that in the specific case as illustrated in Fig. 3(c), the SAM value of HDGMamba is marginally higher than that of RoFDiff. However, its clear and substantial advantages across all other metrics establish the overall superior performance of HDGMamba.

To further investigate this scenario, Fig. 4 provides visual results for Case 6. In Fig. 4(a)–(d), (h), (i), and (j), severe

TABLE II
QUANTITATIVE ASSESSMENT OF UNTRAINED NOISE ON PAVIA CENTER DATASET

Models	Case 1					Case 2				
	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑
CNN-FUS [16]	33.642	0.992	9.167	1.002	0.967	32.467	0.980	13.197	1.226	0.950
SST [24]+PSRT [38]	38.906	0.990	3.714	0.555	0.990	38.589	0.989	3.879	0.571	0.989
SERT [25]+PSRT	40.556	0.993	2.983	0.451	0.993	40.286	0.993	3.080	0.466	0.992
Hylora [26]+PSRT	39.882	0.992	3.435	0.484	0.992	39.667	0.992	3.561	0.493	0.992
SST+DCTFormer [40]	39.037	0.992	3.326	0.498	0.993	39.045	0.992	3.421	0.500	0.993
SERT+DCTFormer	40.644	0.994	2.958	0.418	0.994	40.493	0.994	3.037	0.425	0.994
Hylora+DCTFormer	40.113	0.994	3.213	0.446	0.994	39.991	0.993	3.311	0.450	0.993
SST+PMamba [22]	38.835	0.991	4.641	0.525	0.991	37.785	0.990	5.056	0.582	0.990
SERT+PMamba	40.373	0.994	3.593	0.436	0.993	39.086	0.993	3.859	0.495	0.993
Hylora+PMamba	39.406	0.992	4.333	0.487	0.992	38.371	0.991	4.646	0.540	0.991
SST+FMamba [23]	41.319	0.995	3.276	0.400	0.995	41.101	0.995	3.394	0.409	0.994
SERT+FMamba	42.461	**0.996**	2.730	0.349	**0.996**	42.301	**0.996**	2.801	0.355	**0.996**
Hylora+FMamba	41.828	0.995	3.088	0.374	0.995	41.673	0.995	3.174	0.380	0.995
RoFDiff [42]	41.929	**0.996**	**2.290**	0.349	0.995	41.925	**0.996**	**2.292**	0.350	**0.996**
AHMNet [53]	42.814	**0.996**	2.501	0.338	**0.996**	**42.796**	**0.996**	2.522	0.348	**0.996**
HDGMamba	**42.901**	**0.996**	2.497	**0.334**	**0.996**	42.766	**0.996**	2.514	**0.340**	**0.996**
Models	Case 3					Case 4				
	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑
CNN-FUS [16]	31.830	0.992	10.550	1.855	0.930	32.146	0.990	19.204	3.038	0.915
SST [24]+PSRT [38]	39.322	0.991	3.285	0.530	0.990	37.090	0.986	5.115	0.671	0.985
SERT [25]+PSRT	40.623	0.993	2.826	0.450	0.993	39.266	0.991	3.846	0.517	0.991
Hylora [26]+PSRT	40.181	0.993	3.152	0.469	0.992	38.725	0.990	4.340	0.548	0.990
SST+DCTFormer [40]	39.410	0.992	3.100	0.480	0.993	38.164	0.991	4.029	0.552	0.991
SERT+DCTFormer	40.684	0.994	2.898	0.417	0.994	39.883	0.993	3.578	0.459	0.993
Hylora+DCTFormer	40.245	0.994	3.051	0.437	0.994	39.371	0.992	3.904	0.489	0.993
SST+PMamba [22]	39.463	0.993	3.914	0.485	0.992	35.847	0.984	6.822	0.732	0.984
SERT+PMamba	40.346	0.994	3.335	0.433	0.994	37.609	0.990	5.053	0.590	0.990
Hylora+PMamba	39.817	0.993	3.800	0.461	0.993	37.053	0.988	5.781	0.635	0.988
SST+FMamba [23]	41.773	0.995	2.958	0.380	0.995	39.859	0.993	4.163	0.471	0.992
SERT+FMamba	42.603	**0.996**	2.632	0.344	**0.996**	41.558	0.995	3.282	0.387	0.995
Hylora+FMamba	42.177	**0.996**	2.888	0.360	0.995	40.971	0.994	3.678	0.414	0.994
RoFDiff [42]	41.853	**0.996**	**2.335**	0.351	0.995	41.892	**0.996**	**2.315**	0.350	0.995
AHMNet [53]	42.615	**0.996**	2.587	0.347	**0.996**	42.461	**0.996**	2.658	0.351	**0.996**
HDGMamba	**42.895**	**0.996**	2.489	**0.335**	**0.996**	**42.589**	**0.996**	2.562	**0.349**	**0.996**
Models	Case 5					Case 6				
	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑
CNN-FUS [16]	31.624	0.989	20.313	3.125	0.901	29.864	0.977	24.118	3.258	0.885
SST [24]+PSRT [38]	36.151	0.983	5.863	0.740	0.982	29.856	0.935	12.985	1.493	0.934
SERT [25]+PSRT	38.620	0.990	4.312	0.553	0.990	34.311	0.976	7.983	0.882	0.975
Hylora [26]+PSRT	38.278	0.990	4.574	0.574	0.989	34.825	0.979	7.382	0.829	0.978
SST+DCTFormer [40]	37.611	0.989	4.440	0.590	0.990	34.718	0.979	7.235	0.821	0.980
SERT+DCTFormer	39.497	0.993	3.711	0.478	0.992	37.252	0.987	5.372	0.616	0.987
Hylora+DCTFormer	39.053	0.992	4.111	0.505	0.992	36.725	0.987	5.573	0.649	0.987
SST+PMamba [22]	33.976	0.977	8.217	0.903	0.978	27.369	0.891	18.448	1.969	0.891
SERT+PMamba	35.692	0.986	5.952	0.726	0.988	31.791	0.961	11.404	1.153	0.961
Hylora+PMamba	35.525	0.985	6.381	0.743	0.987	32.337	0.966	10.712	1.082	0.966
SST+FMamba [23]	38.983	0.991	4.709	0.520	0.991	32.254	0.961	10.504	1.123	0.959
SERT+FMamba	40.973	0.995	3.596	0.412	0.994	36.489	0.985	6.465	0.680	0.984
Hylora+FMamba	40.645	0.994	3.822	0.428	0.994	36.795	0.986	6.174	0.654	0.985
RoFDiff [42]	41.831	**0.996**	**2.308**	**0.352**	0.995	39.487	**0.993**	**3.524**	**0.429**	**0.993**
AHMNet [53]	**42.315**	**0.996**	2.580	0.371	**0.996**	38.512	0.992	3.755	0.491	0.992
HDGMamba	42.259	**0.996**	2.605	0.363	**0.996**	**39.530**	**0.993**	3.702	0.486	0.992

quality degradation is observed, with the enlarged red and yellow boxes highlighting various artifacts caused by Gaussian noise, deadline noise, impulse noise, and stripe noise. For most compared methods, stripe noise is the predominant source of spatial distortion, leading to significant declines in fusion quality. In addition, the SAM error maps expose substantial spectral distortions in these methods. Conversely, the visualization results of the proposed HDGMamba closely resemble the GT image. Although minor stripe noise remains visible in Fig. 4(p), the preserved spatial textures and structures confirm high spatial fidelity, while the corresponding SAM error map reflects strong spectral fidelity. In conclusion, HDGMamba consistently delivers exceptional performance across diverse datasets, validating its generalization capability.

3) Comparison Results of WDC Dataset: The WDC dataset possesses richer spectral information than the Pavia and Chikusei datasets, attributable to its increased number of spectral channels, thereby imposing greater demands on model

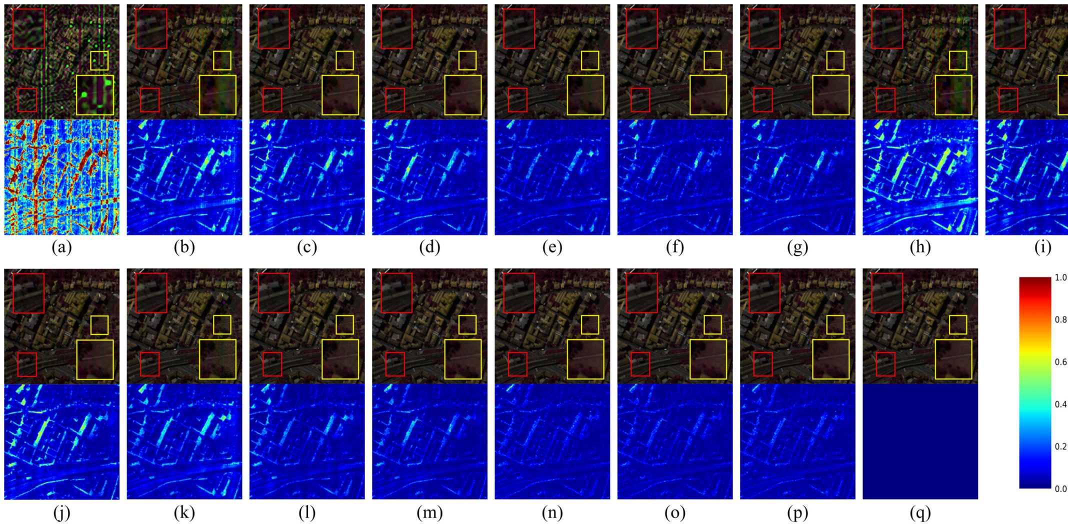

Fig. 2. Visual results of Case 6 on the Pavia Center dataset. (a) CNN-FUS, (b) SST + PSRT, (c) SERT + PSRT, (d) Hylora + PSRT, (e) SST + DCTFormer, (f) SERT + DCTFormer, (g) Hylora + DCTFormer, (h) SST + PMamba, (i) SERT + PMamba, (j) Hylora + PMamba, (k) SST + FMamba, (l) SERT + FMamba, (m) Hylora + FMamba, (n) RoFDiff, (o) AHMNet, (p) HDGMamba, and (q) GT. The odd-numbered rows and the even-numbered rows are the predicted HRHS images and the SAM error maps for each method, respectively, where the RGB images are shown by utilizing the 20th, 40th, and 60th bands of original HS images, and the SAM error maps ranging from 0 to 1 are calculated from all bands.

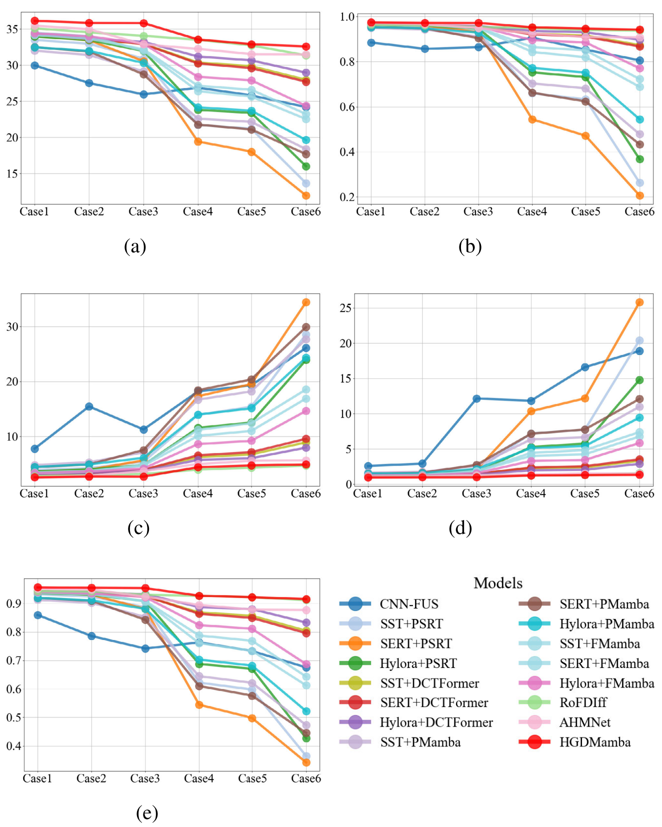

Fig. 3. Results of the Chikusei dataset on untrained noises. (a) PSNR. (b) SSIM. (c) SAM. (d) ERGAS. (e) CC.

performance. To comprehensively assess generalization capability, we rigorously evaluate all compared methods on the WDC dataset. As quantitatively summarized in Table IV under noisy training conditions, the proposed HDGMamba delivers superior fused image quality across most metrics. Although HDGMamba achieves the second-best results in PSNR and ERGAS, it ranks first in all remaining metrics. Specifically, its PSNR is 0.442 dB lower than that of AHMNet, and its ERGAS is 0.006 higher than that of Hylora + DCFormer.

TABLE III
QUANTITATIVE ASSESSMENT OF TRAINED NOISE ON CHIKUSEI DATASET

Models	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑
CNN-FUS [16]	32.854	0.925	5.326	1.694	0.914
SST [24] +PSRT [35]	35.228	0.972	2.936	1.076	0.952
SERT [25]+PSRT	35.568	0.973	2.728	1.027	0.954
Hylora [26]+PSRT	35.482	0.973	2.788	1.039	0.953
SST+DCTFormer [37]	34.816	0.969	2.752	1.083	0.950
SERT+DCTFormer	34.865	0.969	2.712	1.072	0.950
Hylora+DCTFormer	34.763	0.969	2.751	1.085	0.949
SST+PMamba [22]	34.078	0.965	3.277	1.200	0.942
SERT+PMamba	34.315	0.966	3.082	1.160	0.943
Hylora+PMamba	34.222	0.966	3.172	1.171	0.943
SST+FMamba [23]	35.482	0.971	2.693	1.031	0.952
SERT+FMamba	35.656	0.972	2.586	1.002	0.953
Hylora+FMamba	35.572	0.972	2.642	1.014	0.952
RoFDiff [42]	35.126	0.964	2.782	1.062	0.943
AHMNet [53]	35.748	0.973	2.423	1.002	0.955
HDGMamba	**36.555**	**0.977**	**2.359**	**0.924**	**0.961**

TABLE IV
QUANTITATIVE ASSESSMENT OF TRAINED NOISE ON WDC DATASET

Models	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑
CNN-FUS [16]	29.154	0.997	4.922	3.587	0.878
SST [24] +PSRT [35]	32.134	0.994	2.621	2.029	0.949
SERT [25]+PSRT	31.472	0.994	2.174	4.150	0.907
Hylora [26]+PSRT	33.123	0.996	2.132	2.292	0.945
SST+DCTFormer [37]	33.740	0.995	2.521	1.011	0.972
SERT+DCTFormer	36.827	**0.998**	1.739	0.827	0.982
Hylora+DCTFormer	36.533	0.997	1.854	**0.735**	0.984
SST+PMamba [22]	31.226	0.993	2.866	2.358	0.939
SERT+PMamba	30.125	0.993	2.347	5.325	0.884
Hylora+PMamba	32.009	0.995	2.293	2.912	0.932
SST+FMamba [23]	33.074	0.995	2.550	1.689	0.953
SERT+FMamba	33.045	0.996	2.008	3.059	0.925
Hylora+FMamba	34.254	0.997	2.040	1.901	0.954
RoFDiff [42]	33.718	0.993	2.628	1.307	0.969
AHMNet [53]	**37.473**	**0.998**	1.641	0.784	**0.986**
HDGMamba	37.031	**0.998**	**1.619**	0.741	**0.986**

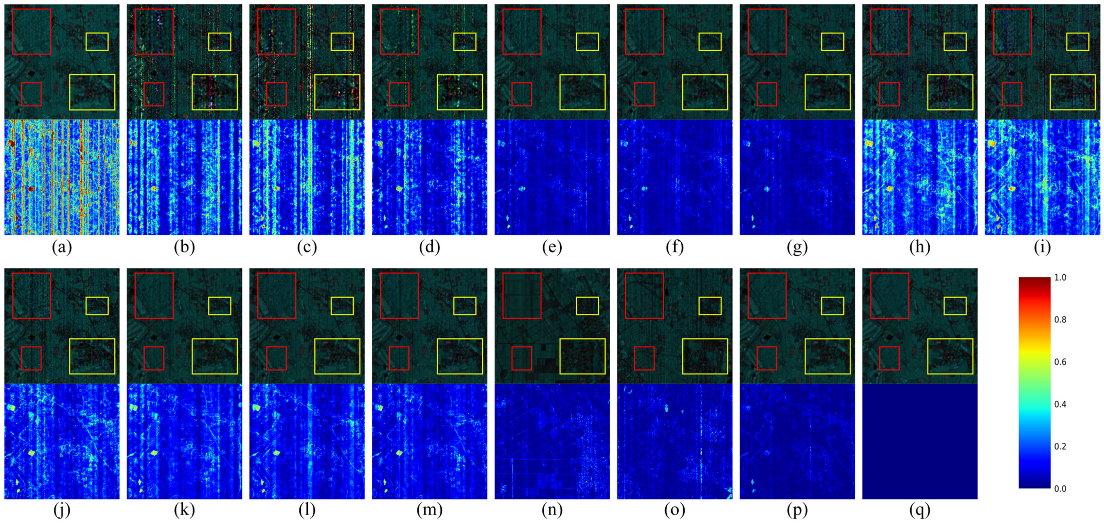

Fig. 4. Visual results of Case 6 in the Chikusei dataset. (a) CNN-FUS, (b) SST + PSRT, (c) SERT + PSRT, (d) Hylora + PSRT, (e) SST + DCTFormer, (f) SERT + DCTFormer, (g) Hylora + DCTFormer, (h) SST + PMamba, (i) SERT + PMamba, (j) Hylora + PMamba, (k) SST + FMamba, (l) SERT + FMamba, (m) Hylora + FMamba, (n) RoFDiff, (o) AHMNet, (p) HDGMamba, and (q) GT. The RGB images are shown by utilizing the 40th, 80th, and 100th bands of the original HS images.

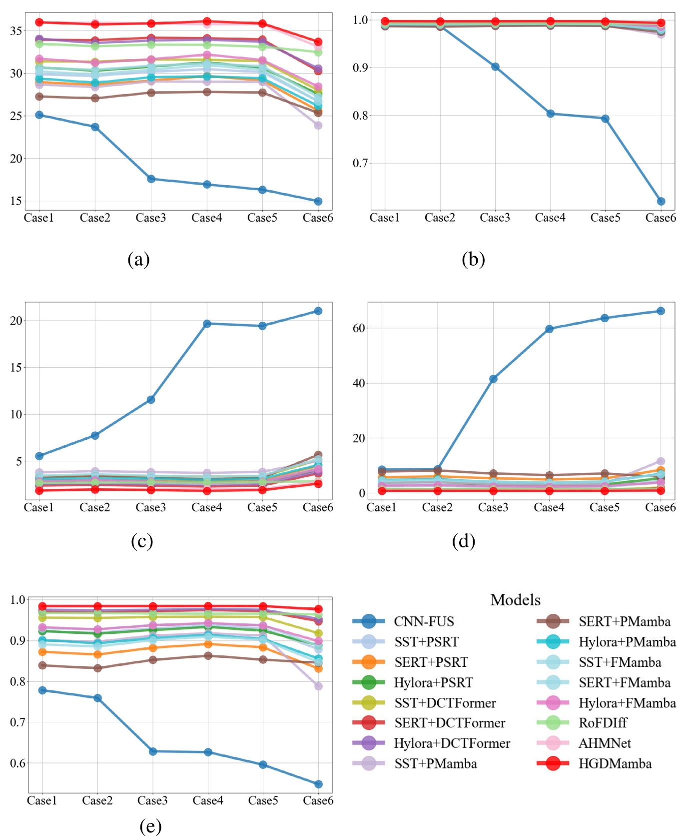

Fig. 5. Results of WDC on untrained noises. (a) PSNR. (b) SSIM. (c) SAM. (d) ERGAS. (e) CC.

More importantly, the visual comparisons from Cases 1–6 as illustrated in Fig. 5 consistently confirm the optimal performance of HDGMamba. Specifically, Fig. 5(a) and (d) demonstrates that HDGMamba attains the highest PSNR and the lowest ERGAS across all untrained noise types when compared to other methods. These quantitative and visual outcomes jointly verify that HDGMamba maintains superior spatial and spectral fidelity under diverse and unknown noise conditions.

Complementing the quantitative assessment, Fig. 6 presents visual fusion results alongside corresponding SAM error maps for all evaluated methods under Case 6. Fig. 6(a) reveals distinct color distortion within enlarged red and yellow bounding boxes, confirming spectral degradation. Moreover, visual results from other comparative methods reveal varying degrees of spatial distortion, predominantly manifesting as black speckle noise, notably evident in Fig. 6(e), (h), (i), (j), and (k). In contrast, the result of HDGMamba shows clearly edges and textures that closely match the ground truth, demonstrating excellent spatial fidelity. Besides, the comparison of SAM error maps further confirms the optimal spectral fidelity of HDGMamba. Compared with others, the SAM error map of HDGMamba exhibits predominantly deeper blue, indicating smaller spectral angle errors.

In conclusion, the proposed HDGMamba delivers exceptional spatial and spectral fidelity on the WDC dataset, ultimately demonstrating robust generalization capabilities beyond these compared methods.

4) Comparison Results on Tuscany Dataset: To evaluate the performance under real-world noise conditions, we employ the Tuscany dataset containing authentic noise. The proposed HDGMamba achieves the best values in PSNR, SSIM, ERGAS, and CC, as shown in Table V. Although its SAM value is slightly higher than that of RoFDiff by 0.019, ranking second, it demonstrates clear advantages in all other metrics. Specifically, PSNR, SSIM, and CC are improved by 0.888, 0.003, and 0.004 dB, respectively, while ERGAS is reduced by 0.063. Furthermore, HDGMamba exceeds AHMNet, the second-best method, with gains of 0.071 dB in PSNR and 0.001 in CC, and reductions of 0.051 in SAM and 0.023 in ERGAS.

In addition to quantitative assessment, a qualitative analysis is provided in Fig. 7 through fused images and their corresponding SAM error maps. To facilitate detailed visual

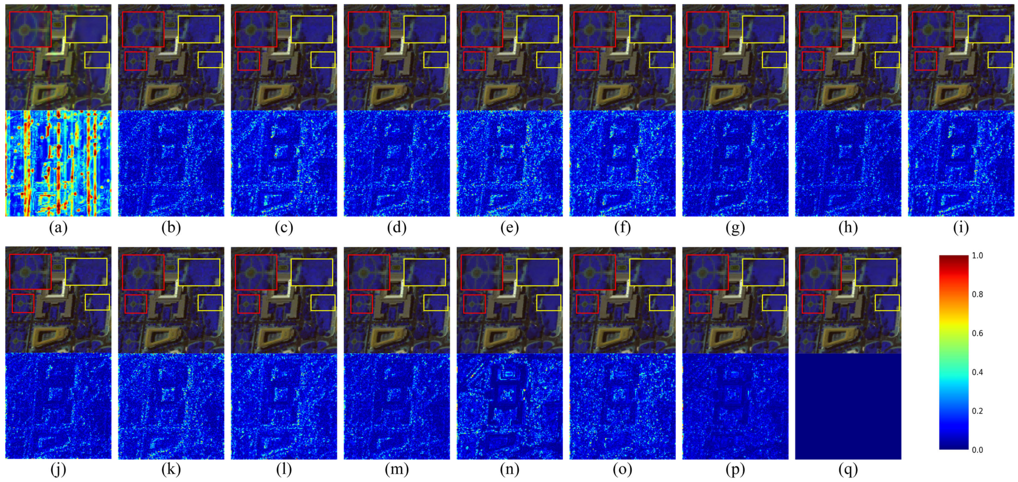

Fig. 6. Visual results of Case 6 in the WDC dataset. (a) CNN-FUS, (b) SST + PSRT, (c) SERT + PSRT, (d) Hylora + PSRT, (e) SST + DCTFormer, (f) SERT + DCTFormer, (g) Hylora + DCTFormer, (h) SST + PMamba, (i) SERT + PMamba, (j) Hylora + PMamba, (k) SST + FMamba, (l) SERT + FMamba, (m) Hylora + FMamba, (n) RoFDiff, (o) AHMNet, (p) HDGMamba, and (q) GT. The RGB images are shown by utilizing the 20th, 40th, and 60th bands of the original HS images.

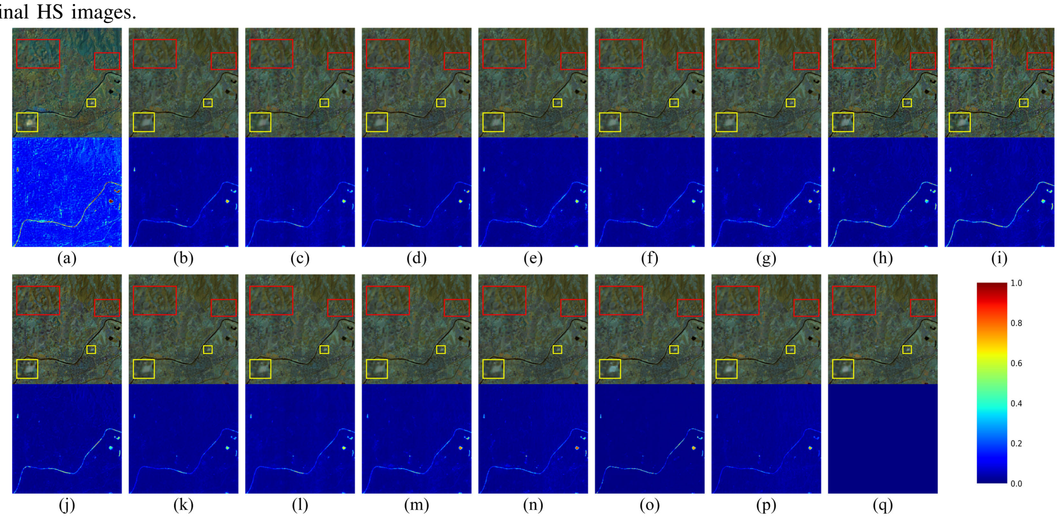

Fig. 7. Visual results in the Tuscany dataset. (a) CNN-FUS, (b) SST + PSRT, (c) SERT + PSRT, (d) Hylora + PSRT, (e) SST + DCTFormer, (f) SERT + DCTFormer, (g) Hylora + DCTFormer, (h) SST + PMamba, (i) SERT + PMamba, (j) Hylora + PMamba, (k) SST + FMamba, (l) SERT + FMamba, (m) Hylora + FMamba, (n) RoFDiff, (o) AHMNet, (p) HDGMamba, and (q) GT. The RGB images are shown by utilizing the 40th, 80th, and 120th bands of the original HS images.

TABLE V
QUANTITATIVE ASSESSMENT OF REAL-WORLD NOISE ON TUSCANY DATASET

Models	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑
CNN-FUS [16]	26.135	0.835	9.229	2.229	0.857
SST [24]+PSRT [38]	35.253	0.969	3.024	0.784	0.957
SERT [25]+PSRT	31.937	0.935	4.662	1.261	0.915
Hylora [26]+PSRT	35.306	0.969	3.016	0.796	0.958
SST+DCTFormer [40]	36.274	0.972	2.896	0.629	0.966
SERT+DCTFormer	36.265	0.970	3.112	0.666	0.962
Hylora+DCTFormer	36.898	0.974	2.664	0.582	0.968
SST+PMamba [22]	31.472	0.939	4.523	1.094	0.939
SERT+PMamba	29.306	0.904	6.139	1.675	0.907
Hylora+PMamba	31.505	0.941	4.377	1.089	0.943
SST+FMamba [23]	34.672	0.963	3.122	0.725	0.953
SERT+FMamba	34.038	0.957	3.575	0.929	0.943
Hylora+FMamba	35.871	0.970	2.682	0.648	0.962
RoFDiff [42]	39.083	0.986	**1.794**	0.447	0.983
AHMNet [53]	39.900	**0.989**	1.864	0.407	0.986
HDGMamba	**39.971**	**0.989**	1.813	**0.384**	**0.987**

comparison, enlarged regions within red and yellow boxes highlight topographic and village details, respectively. Visual inspection reveals severe distortion in the red-box areas of Fig. 7(a), (c), (h), (i), and (j). The yellow-box regions in Fig. 7(a), (d), (f), (l), (m), and (o) exhibits varying degrees of color distortion. Moreover, Fig. 7(b), (c), (e), (h), (i), and (j) suffers from both color distortion and blurred edges. A comparison of the SAM error maps further indicates that all compared methods exhibit some level of spectral distortion, which is particularly pronounced in river areas. In contrast, the SAM error map of HDGMamba shows the least spectral distortion in river regions and appears deeper blue overall, indicating the lower spectral error across the entire scene. In summary, both quantitative and qualitative results demonstrate that the proposed HDGMamba delivers satisfactory fusion performance even under real-world noisy conditions.

C. Ablation Study

1) Effectiveness of HDB Components: To evaluate the contribution of the core components in the HDB,

TABLE VI
PROGRESSIVE MODULE ABLATION IN HDB

Models	LDF	HM-SSM	HGM	PSNR↑	SSIM↑	SAM↓	ERGAS↓	CC↑
HDGMamba-1	×	×	×	39.983	0.994	3.226	0.452	0.993
HDGMamba-2	✓	×	×	42.017	0.996	2.612	0.359	0.995
HDGMamba-3	✓	×	✓	43.857	**0.997**	2.220	0.318	**0.997**
HDGMamba-4	✓	✓	×	43.645	0.996	2.324	0.317	0.995
HDGMamba	✓	✓	✓	**44.434**	**0.997**	**2.109**	**0.288**	**0.997**

HDGMamba-1, HDGMamba-2, HDGMamba-3, and HDGMamba-4 are designed to be compared. Specifically, HDGMamba-1 replaces the LDF, HM-SSM, and HGM with convolutional layers to directly learn the nonlinear mapping from noisy LRHSI and HRMSI to HRHSI. HDGMamba-2 incorporates the LDF to denoise the LRHSI. HDGMamba-3 further adds the HGM to remove noise from the HRMSI. HDGMamba-4 builds on HDGMamba-2 by integrating the HM-SSM to fuse spectral and spatial features.

As shown in Table VI, HDGMamba achieves the best performance. HDGMamba-1, which lacks all core components of the HDB, performs the worst due to its limited ability to suppress noise. Compared with HDGMamba-1, the HDGMamba-2 improves PSNR, SSIM, and CC by 2.034, 0.002, and 0.002 dB, respectively, while reducing SAM and ERGAS by 0.614 and 0.066. This confirms the effectiveness of the LDF. By introducing the HGM to remove the noise of HRMSI, HDGMamba-3 achieves gains across all metrics compared to HDGMamba-2, confirming the importance of HGM for the fusion quality. The HM-SSM also contributes substantially, as evidenced by the performance of HDGMamba-4, which surpasses HDGMamba-2 by raising PSNR by 1.628 dB and lowering SAM and ERGAS by 0.288 and 0.042, respectively. However, since the HM-SSM operates on the noisy HRMSI, some fusion degradation occurs. Compared with HDGMamba-3, HDGMamba-4 exhibits decreases in PSNR, SSIM, and CC of 0.212, 0.001, and 0.002 dB, respectively, along with increases in SAM and ERGAS of 0.104 and 0.001, respectively.

In contrast, HDGMamba combines the LDF, HM-SSM, and HGM to achieve superior results. Compared to HDGMamba-4, HDGMamba improves PSNR, SSIM, and CC by 0.789, 0.001, and 0.002 dB, respectively, while reducing SAM and ERGAS by 0.215 and 0.082, respectively. Moreover, the HGM effectively suppresses spatial noise and extracts clean high-frequency features, thereby enhancing the state evolution process within the HM-SSM. In summary, the LDF, HM-SSM, and HGM are all critical for reconstructing HRHSI under noisy conditions. Their cooperative interaction significantly improves both the spatial and spectral fidelity of the fused image.

2) Spatial Denoising Method Comparison: To further assess the effectiveness of the proposed HGM, we compare it with several representative spatial denoising modules by replacing HGM with SST, SERT, and Hylora, denoted as HDGMamba-D1, HDGMamba-D2, and HDGMamba-D3, respectively. Unlike the previous ablation study that removes HGM entirely, this experiment focuses on evaluating whether HGM outperforms other existing methods in guiding the HM-SSM via denoised spatial features.

TABLE VII
ABLATION STUDY OF SPATIAL DENOISING

Models	Modules	PSNR↑	SSIM↑	SAM↓	ERGAS↓	CC↑
HDGMamba-D1	SST	43.405	0.996	2.365	0.322	0.995
HDGMamba-D2	SERT	43.826	0.996	2.293	0.306	0.996
HDGMamba-D3	Hylora	43.750	**0.997**	2.314	0.309	0.996
HDGMamba	HGM	**44.434**	**0.997**	**2.109**	**0.288**	**0.997**

TABLE VIII
ABLATION STUDY OF HM-SSM

Models	Modules	PSNR↑	SSIM↑	SAM↓	ERGAS↓	CC↑
HDGMamba-S	SSM	43.920	0.996	2.197	0.303	0.996
HDGMamba	HM-SSM	**44.434**	**0.997**	**2.109**	**0.288**	**0.997**

Table VII shows the comparison results, with HDGMamba-D1 exhibiting the lowest fusion quality among the compared methods, primarily due to the limited denoising capability of SST. Meanwhile, the comparison between HDGMamba-D2 and HDGMamba-D3 reveals that SERT excels in handling global errors, while Hylora focuses more on preserving texture details. For instance, HDGMamba-D2 achieves the second-best performance in PSNR, SAM, and ERGAS, whereas HDGMamba-D3 gains the highest SSIM. However, both denoising methods lead to quality degradation due to their tendency to oversmooth when removing noise, resulting in blurred textures and edges. In contrast, HGM separates noise from high-frequency regions based on their unstructured gradients, thereby preserving fine spatial details. Consequently, the proposed HDGMamba significantly enhances fusion quality and consistently outperforms all compared methods.

3) Effectiveness of Adoptive Timescale Parameters: To validate the adaptive timescale mechanism in HM-SSM, we design variant HDGMamba-S using standard SSM with fixed timescale parameters. Unlike the standard SSM, HM SSM dynamically adjusts timescale parameters guided by high-frequency features. Table VIII compares HDGMamba S and HDGMamba, demonstrating the performance impact of adaptive parameterization.

Compared with the proposed HDGMamba, HDGMamba-S exhibits lower performance, with PSNR, SSIM, and CC reduced by 1.143, 0.001, and 0.001 dB, respectively, and SAM and ERGAS increased by 0.088 and 0.015, respectively. Although the standard SSM dynamically adjusts its parameter matrix based on the input, its fixed time-step parameters are less effective in adapting to variations in

TABLE IX
COMPLEXITY OF DIFFERENT MODELS ON PAVIA CENTER DATASET

Models	Parameters (M)	Flops (G)	Memory (GB)	Inference (S)
CNN-FUS [16]	-	-	-	20.449
SST [24] +PSRT [38]	23.530	397.341	5.140	1.203
SERT [25]+PSRT	9.640	165.908	4.930	1.251
Hylora [26]+PSRT	13.010	225.764	4.704	0.917
SST+DCTFormer [40]	21.050	1448.339	6.248	1.516
SERT+DCTFormer	17.680	1388.483	6.038	1.857
Hylora+DCTFormer	21.050	1448.339	5.812	1.516
SST+PMamba [22]	23.480	393.626	4.942	1.208
SERT+PMamba	9.590	162.193	4.732	1.256
Hylora+PMamba	12.960	222.049	4.506	0.915
SST+FMamba [23]	25.380	417.241	4.862	3.212
SERT+FMamba	11.490	185.808	4.652	3.266
Hylora+FMamba	14.860	245.664	4.426	2.925
RoFDiff [42]	92.690	3235.890	4.958	0.719
AHMNet [53]	**0.230**	**21.632**	**1.078**	0.251
HDGMamba	1.290	84.007	1.188	**0.156**

complex and flat texture regions, leading to reduced quality in detailed textures. In contrast, the HM-SSM utilizes the clean high-frequency features from the HGM to adjust its time-step parameters, adaptively guiding dynamic modeling to enhance texture details, thereby significantly improving image quality.

D. Comparison of Computational Complexity

We evaluate the computational efficiency of various methods on the Pavia Center dataset by measuring their FLOPs, number of parameters, memory usage, and inference time. As shown in Table IX, CNN-FUS requires multiple iterations to refine HRHSIs, leading to slow inference speed. The Transformer-based methods (SST + PSRT, SERT + PSRT, Hylora + PSRT, SST + DCTFormer, SERT + DCTFormer, and Hylora + DCTFormer) exhibit high computational complexity, primarily because their attention mechanisms require pairwise calculations across all pixels, significantly increasing FLOPs and memory usage. Although SST + PSRT, SERT + PSRT, and Hylora + PSRT adopt window attention to reduce FLOPs, their complexities remain higher than those of Mamba-based methods (SST + PMamba, SERT + PMamba, Hylora + PMamba, SST + FMamba, SERT + FMamba, Hylora + FMamba, and AHMNet). In contrast, Mamba-based methods employ selective recursive scanning via SSM, which focus on key information with linear computational complexity, thereby substantially reducing the cost in terms of FLOPs and memory.

Among Mamba-based methods, SST + PMamba, SERT + PMamba, Hylora + PMamba, SST + FMamba, SERT + FMamba, and Hylora + FMamba require an additional denoising network to obtain clean LRHSI and HRMSI inputs, which leads to a significant increase in FLOPs, parameters, memory, and inference time. Interestingly, RoFDiff, despite having the highest parameter count and FLOPs due to its complex architecture and iterative inference process, achieves a substantially reduced inference time. This is because its inherent noise-resistant fusion capability avoids the need for a separate denoising stage, demonstrating that robust MHIF methods can achieve satisfactory fusion outcomes with lower practical computational cost than the cascaded denoising-fusion method.

It is noteworthy that the proposed HDGMamba achieves a faster inference time (by 0.095 s) compared to AHMNet, despite having higher FLOPs, parameter count, and memory usage. This efficiency gain primarily stems from the additional computational overhead in AHMNet, which employs a low-rank subspace combined with DWT for denoising. When considering its robust noise-resistant fusion performance, HDGMamba effectively balances model complexity with practical inference speed, demonstrating a more favorable tradeoff between effectiveness and computational cost.

E. Generalization to Other Vision Tasks

To assess the broader applicability of the combined HGM and HM-SSM framework, the proposed HDGMamba is extended to the HSI denoising and pansharpening tasks, respectively.

1) Denoising: Experiments are conducted using the public Interdisciplinary Computational Vision Laboratory (ICVL) dataset. For training, 100 HSIs are selected, each with a spatial size of 1392 × 1300, 1392 × 1300 pixels and containing 31 spectral bands. These images are divided into 64 × 64 × 31 patches and augmented through random flipping, cropping, and resizing. The independent test set consists of 50 images with dimensions of 512 × 512 × 31.

Both the Gaussian and mixture noise settings are adopted for evaluation. The Gaussian noise, where the standard deviation α is sampled from the ranges [0, 15], [0, 55], and [0, 95] to represent low, medium, and high noise intensities, respectively. The mixture noise composed of four components: 1) Gaussian noise with $\alpha \in [0, 95]$; 2) impulse noise applied to one-third of the spectral bands, with corruption levels between 10% and 70%; 3) stripe noise affecting 5%–15% of the columns in one-third of the bands; and 4) deadline noise affecting 5%–15% of the columns in another one-third of the bands.

The original HDGMamba model is designed for MHIF under noisy conditions. For the denoising task, a noisy HSI is first separated into high-frequency and low-frequency components. The LDF and HGM are employed to remove noise of low-frequency and high-frequency features, respectively, and the HM-SSM then fuses these denoised features. As shown in Table X, the proposed HDGMamba is compared to seven DL-based HSI denoising methods, including trainable spectral–spatial sparse coding model (T3SC) [54], SST [24], SERT [25], spatial–spectral UMamba (SSUMamba) [55], and linear attention mamba (LaMamba) [56]. The HDGMamba achieves the best values across all metrics under low Gaussian noise conditions (α = [0, 15]). However, its performance degrades as noise intensity increases. Under the mixture noise setting, HDGMamba shows a reduction of 1.619 dB in PSNR and 0.012 in SSIM compared to LaMamba, while its SAM value increases by 0.014. This performance is due to the design of the HGM module, which is optimized to extract clean high-frequency details from images (HRMS images) with relatively lower noise levels. When applied to HSI denoising tasks with strong and complex noise patterns, the high-frequency features recovered by the HGM can retain noise residuals, limiting the overall performance.

TABLE X
QUANTITATIVE COMPARISON OF HYPERSPECTRAL DENOISING ON ICVL DATASET

Models	α=[0,15]			α=[0,55]		
	PSNR ↑	SSIM ↑	SAM ↓	PSNR ↑	SSIM ↑	SAM ↓
T3SC [54]	50.910	0.994	0.027	45.890	0.986	0.049
SST [24]	49.230	0.993	0.031	46.460	0.987	0.035
SERT [25]	50.300	0.993	0.025	46.410	0.988	0.034
SSUMamba [55]	51.830	**0.995**	0.023	47.610	0.990	0.032
LaMamba [56]	52.370	**0.995**	**0.021**	**47.990**	**0.991**	**0.031**
HDGMamba	**2.377**	**0.995**	**0.021**	47.730	**0.991**	0.032
Models	α=[0,95]			Mixture		
	PSNR ↑	SSIM ↑	SAM ↓	PSNR ↑	SSIM ↑	SAM ↓
T3SC [54]	43.200	0.977	0.067	36.550	0.925	0.176
SST [24]	44.640	0.983	0.041	41.040	0.974	0.063
SERT [25]	44.790	0.984	0.039	40.030	0.957	0.067
SSUMamba [55]	45.680	**0.986**	0.039	44.130	0.978	0.052
LaMamba [56]	**45.990**	**0.986**	**0.037**	**44.570**	**0.981**	**0.049**
HDGMamba	45.510	0.985	0.041	42.951	0.969	0.063

TABLE XI
QUANTITATIVE COMPARISON OF PANSHARPENING ON WV3 DATASET

Models	SAM ↓	ERGAS ↓	Q8 ↑
PNN [34]	3.680	2.682	0.893
DCFNet [58]	3.038	2.165	0.913
LAGConv [59]	3.104	2.300	0.910
HMPNet [60]	3.063	2.229	0.916
CMT [61]	2.994	2.214	0.917
CANNet [62]	2.930	2.158	0.920
ARNet [63]	**2.885**	2.139	**0.921**
HDGMamba	2.903	**2.135**	0.919

2) Pansharpening: The WorldView-3 (WV3) dataset, containing eight-band multispectral images and corresponding panchromatic images, is used to evaluate the pansharpening task. Following Wald's protocol [57], the training set comprises thousands of image triplets with panchromatic (PAN) image ($64 \times 64 \times 1$), LRMSI ($64 \times 64 \times 8$), and ground truth ($64 \times 64 \times 8$) patches. The test set includes 20 such triplets, all with a spatial size of $256 \times 256 \times 1$ pixels. Performance is assessed via SAM, ERGAS, and the eight-band quality index (Q8), where Q8 measures joint spectral–spatial fidelity.

DL-based pansharpening methods, including pansharpening by CNN (PNN) [34], dynamic cross feature fusion network (DCFNet) [58], local-context adaptive convolution kernels (LAGConvs) [59], HMPNet (interpretable model-driven deep network for hyperspectral, multispectral, and panchromatic image fusion) [60], (cross modulation transformer (CMT) with hybrid loss for pansharpening) [61], content-adaptive nonlocal convolution network (CANNet) [62], and adaptive rectangular convolution (ARNet) [63] are used for comparisons. As shown in Table XI, the proposed HDGMamba achieves the best ERGAS, although it shows a slight increase of 0.018 in SAM and a slight decrease of 0.002 in Q8 compared to ARNet. This performance difference originates from the design objective of HDGMamba, which is optimized for image fusion under noisy conditions. Specifically, the LDF is designed to restore clean spectral information from a noisy LRHSI, while the HGM aims to extract high-frequency details from a noisy HRMSI. However, in the pansharpening task where inputs are predominantly clean, the strong denoising operation of the LDF may lead to an over-smoothing of the spectral features. Simultaneously, the high-frequency features by the HGM, when applied to a clean PAN image, might lose some fine spatial details that are crucial for optimal reconstruction. Consequently, metrics highly sensitive to SAM and Q8 are slightly impacted. In contrast, ERGAS, which measures the global error, is less affected by this localized over-smoothing, allowing HDGMamba to achieve the best result.

In summary, the proposed HDGMamba demonstrates cross-task generalization in both denoising and pansharpening. While it excels at denoising under low-noise intensities, its performance in high-noise scenarios and on clean pansharpening inputs indicates room for improvement. Future work may focus on dynamically adapting its denoising strength based on input characteristics to enhance its robustness across diverse tasks.

V. CONCLUSION

This article introduces the HDGMamba for robust MHIF. To enhance spatial texture details, we propose an HM-SSM that leverages clean high-frequency features to guide the fusion of spatial and spectral information. In addition, to extract these features, the HGM is designed to suppress noise by exploiting structural differences between noise and high-frequency components, rather than relying on traditional smoothing. Extensive experimental results demonstrate that HDGMamba outperforms state-of-the-art methods in the MHIF task under noisy conditions, and ablation studies confirm the effectiveness of these proposed modules. In future work, we will focus on semi-supervised and unsupervised MHIF methods in complex noisy environments.

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Jiangtao Huang received the M.S. degree from the Faculty of Information Science and Engineering, Ningbo University, Ningbo, China, in 2022, where he is currently pursuing the Ph.D. degree.

His research interests include remote sensing image processing.

Ting Luo (Member, IEEE) received the Ph.D. degree from the Faculty of Information Science and Engineering, Ningbo University, Ningbo, China, in 2016.

He is currently a Professor with the College of Science and Technology, Ningbo University. His research interests include multimedia security, image processing, data hiding, pattern recognition, and remote sensing image processing.

Zhouyan He (Member, IEEE) received the Ph.D. degree from the Faculty of Information Science and Engineering, Ningbo University, Ningbo, China, in 2021.

She is currently an Associate Professor with the College of Science and Technology, Ningbo University. Her research interests include image processing, multimedia security, quality assessment, 3-D point cloud processing, and deep learning.

Haiyong Xu (Member, IEEE) received the B.S. degree in applied mathematics from Jilin University, Changchun, China, in 2003, the M.S. degree in applied mathematics from Sun Yat-sen University, Guangzhou, China, in 2005, and the Ph.D. degree from Ningbo University, Ningbo, China, in 2020.

He is currently an Associate Professor with the School of Mathematics and Statistics, Ningbo University. His research interests include underwater image processing and deep learning.

Yang Song received the M.S. and Ph.D. degrees from Ningbo University, Ningbo, China, in 2015 and 2018, respectively.

He is currently an Associate Professor with the College of Science and Technology, Ningbo University. His research interests include multimedia communication, image processing, and visual quality assessment.

Leyuan Fang (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015.

From August 2016 to September 2017, he was a Post-Doctoral Researcher with the Department of Biomedical Engineering, Duke University, Durham, NC, USA. He is currently a Professor with the College of Electrical and Information Engineering, Hunan University. His research interests include sparse representation and multiresolution analysis in remote sensing and medical image processing.

Dr. Fang was a recipient of the One Second-Grade National Award from the Nature and Science Progress of China in 2019. He is an Associate Editor of IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Neurocomputing*.